D03 · Maintenance & Asset Reliability

04 · MRO inventory & critical spares execution

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1 WHAT IT IS

MRO (Maintenance, Repair, Operations) inventory management is the discipline of segmenting and optimizing the spare-parts inventory that supports equipment reliability — distinguishing critical spares (high consequence of stockout), insurance spares (low probability of need but catastrophic if needed), commodity MRO (frequent low-stakes use), and obsolete inventory. The economic model is fundamentally different from production inventory: production inventory is optimized to minimize carrying cost subject to service-level constraint; MRO inventory is optimized to minimize the expected cost of stockout × probability of failure × downtime cost, which produces a very different stocking policy.

2 WHY IT MATTERS

MRO inventory is the most under-managed major working-capital category in manufacturing operations — and the management mistake that produces the largest hidden cost. **The dominant failure mode is treating MRO inventory as a production-inventory cost-minimization problem: applying the same EOQ, safety-stock, and turn-rate metrics that work for finished goods, and arriving at a "right-sized" MRO inventory that produces $300K/day stockouts because the $4K critical bearing was reduced as overstock during the last working-capital review**. The peer-reviewed reference handbook on maintenance management treats this category mistake directly **(Ben-Daya, Ait-Kadi, Knezevic & Raouf, 2009 — peer-reviewed, "Handbook of Maintenance Management and Engineering", Springer)**: MRO inventory optimization is fundamentally a reliability-economics problem, not a working-capital problem, and applying working-capital frameworks systematically destroys availability. Industry data confirms — **well-segmented MRO programs deliver 15–25% total inventory reduction with zero critical stockouts; poorly-segmented programs (cost-minimization-only) typically run 6–12 critical stockouts per year per 100 critical assets, each costing $150K–$500K in lost production**. The math is one-sided: the inventory savings from cutting critical spares are dwarfed by the stockout costs that follow.

3 HOW IT IS DONE

1. **Segment the MRO inventory by criticality × failure-rate × lead-time — not by usage frequency or dollar value.** Critical spares: high consequence of stockout, low to moderate probability of need. Insurance spares: extreme consequence of stockout, very low probability of need, often long lead-time. Commodity MRO: frequent use, low individual stockout consequence. Obsolete: equipment retired, parts no longer needed. Each category has fundamentally different optimization economics; applying one policy across all categories produces wrong stocking decisions in three of four categories.
2. **Calculate criticality from the failure mode the part addresses — not from the part's cost or usage history.** A $4,000 bearing that goes into a critical compressor with no redundancy is a high-criticality spare; a $4,000 motor that goes into a non-critical conveyor with backup is a low-criticality spare. The part-level dollar value tells you nothing about criticality; the consequence of the equipment failure if the part is unavailable tells you everything. Plants that segment by dollar value get the criticality classification wrong half the time and stock the wrong items at high service levels.
3. **For critical spares, stock to availability target — not to cost-minimization target.** The right stocking level for a critical spare is the level that produces the availability target (typically 99%+ for truly critical positions) given the lead time, the failure rate, and the consequence economics. This usually means stocking more than the cost-minimization model recommends, and the math is straightforward: a $30K spare bearing carried for 10 years at 5% holding cost = $15K total carry cost; one stockout event at $400K = wipes out 27 years of carry cost. Plants that resist this math save inventory dollars and lose availability dollars at 20–40x the savings rate.
4. **For commodity MRO, apply traditional inventory optimization — this is where EOQ, safety-stock, and turns metrics work correctly.** The commodity tier (fasteners, lubricants, filters, general consumables) accounts for 70–80% of MRO SKU count but typically 20–30% of MRO dollar value. Standard inventory management techniques work here: EOQ-driven order quantities, statistical safety stock, vendor-managed inventory where supplier reliability supports it, consignment for high-volume low-margin items. The cost-minimization framework that fails on critical spares succeeds here because the stockout consequence is low.

5. **Aggressively identify and dispose of obsolete inventory — the working-capital savings live mostly here, not in cutting critical spares.** Equipment retirement, design changes, supplier discontinuation, and accumulated over-purchasing produce obsolete inventory that typically constitutes 15–25% of MRO dollar value in operations without active management. The disposal is procedurally painful (sunk cost, accounting write-offs, internal politics about who authorized the original purchase) but economically clean — the inventory exists, the equipment does not, the carrying cost is real. Plants that disposition obsolete inventory annually recover the working capital that operations needs without touching critical spares.
6. **Govern MRO inventory through joint reliability + supply chain ownership — not through supply chain alone.** The dominant pathology is supply chain owning MRO inventory under working-capital metrics with reliability engineering consulted but not authoritative. The structural fix is joint ownership: supply chain manages the procurement, storage, and disposition processes; reliability engineering owns the stocking-level decisions for critical and insurance spares; finance applies the right economic framework (carrying cost vs. expected stockout cost) to each segment differently. Plants that establish this joint ownership produce both inventory reduction and availability improvement; plants that maintain supply-chain-only ownership produce inventory reduction and availability destruction.

WORKED EXAMPLE — ILLUSTRATIVE

A pharmaceutical manufacturer (FDA-regulated, 2 plants, 8 production lines) carries **$14M in MRO inventory across ~12,000 SKUs** under a cost-minimization policy applied uniformly. The policy has reduced inventory from $19M to $14M over 30 months, but the operation has experienced **6 critical stockouts in 24 months**, with downtime costs ranging $180K to $420K per event (total documented stockout cost: $1.7M, plus reputational damage with 2 major customers). Re-segmentation using criticality × failure-rate × lead-time: **17 SKUs identified as critical with high failure probability (bearings, seals, control valves on critical positions with 12–18 week lead times); 280 SKUs as insurance spares (extremely low probability of need but catastrophic if needed); 340 SKUs as commodity MRO with optimization potential; 1,840 SKUs as obsolete**. The re-optimization: **critical SKUs inventory increases from $480K to $1.2M (stocked to 99.5% availability target with explicit lead-time and failure-rate calculation); commodity SKUs reduced from $4.8M to $2.1M (traditional EOQ + VMI for top-30 high-volume items); obsolete inventory of $3.2M dispositioned ($820K recovered via supplier credits and resale, $2.4M written off); insurance spares stocked based on consequence economics ($1.4M total)**. Results 14 months after re-optimization: **total MRO inventory $14M -> $10.9M (-22%), critical stockouts 6 in 24 months -> 0 in 14 months, avoided downtime cost estimated at $2.3M, working capital freed = $3.1M**. The cost-minimization-only policy had been simultaneously over-investing in obsolete inventory and under-investing in critical spares; the re-segmentation reversed both.

IN PRACTICE

Segment by criticality × failure-rate × lead-time, not by usage or dollar value

Standard ABC analysis on usage produces the wrong stocking policy — a low-usage, high-criticality, long-lead-time part lands in "C" and gets cut, then takes the production line down for 14 weeks. The MRO segmentation must use criticality and failure-rate and lead-time together, not usage in isolation.

Stock critical spares to availability target, not cost-minimization target

For critical positions, the economic optimum is carrying cost vs (probability of failure × consequence of stockout × downtime cost) **(Ben-Daya, Ait-Kadi, Knezevic & Raouf, 2009 — peer-reviewed, Springer)**. This produces higher stocking levels than production-inventory logic — and the math is one-sided.

Aggressively disposition obsolete inventory — where real working-capital savings live

The MRO inventory reduction opportunity is mostly in obsolete and slow-moving commodity tiers, not in critical spares. Plants that target obsolete inventory typically capture 15–25% inventory reduction with zero availability impact — and free the warehouse space for the critical positions that need to be there.

Joint governance: reliability + supply chain + finance

Supply chain measured on turns; reliability measured on availability — the metrics pull opposite directions on critical positions and supply chain wins because turns show on the P&L while availability shows only when a stockout happens. Joint governance with operations leadership backing reliability's authority on critical spares is the structural prerequisite.

WHAT YOU WOULD USE

MRO INVENTORY MANAGEMENT

SAP Spare Parts Optimization / Oracle Inventory Optimization (with MRO modules) / Servigistics (PTC) — Enterprise platforms with native MRO segmentation logic, criticality-driven stocking calculations, and integration with EAM systems for failure-rate and lead-time data; the right fit for multi-plant operations standardizing MRO management across sites **(SAP — vendor; Oracle — vendor; PTC — vendor)**.

IBM Maximo / Infor EAM (inventory modules with reliability integration) — EAM-integrated inventory management where MRO inventory data lives in the same system as the equipment failure-rate and criticality data; eliminates the data-integration gap that defeats most MRO optimization projects **(IBM — vendor; Infor — vendor)**.

SDI Global / Sourcing partners + Synovos (Vallen) for MRO program management — External MRO program management providers that handle procurement, storage, and disposition as a managed service, with technology platform layered on top; the right fit for operations choosing to outsource MRO operations while retaining strategic stocking decisions **(SDI — vendor; Vallen — vendor)**.

Excel-based segmentation + monthly joint review (supply chain + reliability + finance) + CMMS integration — For operations without specialized platform — structured Excel segmentation by criticality × failure-rate × lead-time, monthly joint review of critical-spare stocking decisions, integration of failure-rate data from CMMS into the segmentation logic. The lowest-tech option produces strong outcomes when the joint governance is real and the segmentation discipline is maintained.

4 SO WHAT

Optimizing MRO inventory by carrying cost — the same logic as production inventory — is saving dollars while losing hundreds of thousands on critical spares that were not there. Segment by criticality × failure-rate × lead-time not by usage or dollar value, calculate criticality from the failure-mode consequence not the part's cost, stock critical spares to availability target not to cost-minimization target, apply traditional optimization to commodity MRO where it works, aggressively disposition obsolete inventory which is where the real working-capital savings live, and govern through joint reliability + supply chain + finance ownership. MRO is a reliability-economics problem treated as a working-capital problem by 80% of operations — and the structural error costs 2–5x more in stockout consequence than it saves in carrying cost.

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